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Multi wavelength optical broad band photometric properties of a representative sample of geostationary satellites

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Abstract

We present the results of multi-wavelength optical observations, using B, V, R and I band filters, of a sample of 61 active geostationary communications satellites, on 2015 July 9–11. These observations were done outside the glinting season, with a longitudinal solar angle smaller than 6° and a latitudinal solar phase angle of $\sim 16.5°$. This approach enabled us to determine the distribution of quiescent magnitudes and colors of these targets during a period of the night when they are expected to be the brightest. We present the distribution of magnitudes and colors of the sample. We find that the colors of these satellites are in general redder than the Solar spectrum, with a wide distribution of values ($0.62 < B-V < 1.49, 1.00 < B-R < 2.28, 1.31 < B-I < 3.07, 0.34 < V-R < 0.90, 0.63 < V-I < 1.69, 0.27 < R-I < 0.79$). Satellites from different manufacturers are found to cluster around specific colors in color-color plots, indicating that this information can be used to identify different satellites and resolve cross tagging issues. We also investigate whether the colors of satellites change with time and find that in most cases there is no significant change to these properties as the satellites age.

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Keywords: Geostationary satellite; Optical photometry; Multi wavelength colors; Spacecraft cross tagging; Material aging

1. Introduction

Determining the multiwavelength properties of a large sample of geostationary satellites (geosats) is an important piece of information needed to understand these objects. Based on this information one can determine the overall properties of these objects, like the typical distribution of magnitudes and colors. This information can be of use in the identification of different satellites (Payne et al., 2002, 2004), and satellite buses, determine the composition of a satellite as a whole, the composition of different components of a satellite (Hall, 2010; Pauca et al., 2003; Rodriguez et al., 2007; Payne et al., 2009), and as a means to diagnose problems. Another important application of this information is in the design of future instruments, such as an interferometer (Schmitt et al., 2011), where this information can be used to optimize performance.

Several efforts (e.g. Sanchez et al., 2000; Payne et al., 2002; Vrba et al., 2003; Cardona et al., 2016; Svendrup et al., 2001; Scott and Wallace, 2009) have collected multiwavelength observations of geosats and GPS satellites. Although the information currently available on these objects is useful for several purposes, sometimes it is limited in the number of objects and may not necessarily have been obtained in a homogeneous way. Here we attempt to overcome some of these issues by obtaining B, V, R and I band images of a large number of geostationary satellites. The different band observations were done within a short period of time and all satellites were observed within a small range of solar phase angle, so as to minimize variability due to changing illumination.

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2. Sample selection and observations

Our initial sample was composed of all 92 active geosynchronous telecommunications and scientific satellites with orbital inclinations $i < 0.5^{\circ}$, located between longitudes of 61°W and 139°W at the time of the observations. This selection criterion was used to ensure the targets were observable from Flagstaff, AZ, with elevations $> 23^{\circ}$ above the horizon, thus avoiding unreasonably large airmasses and atmospheric extinction. This selection method also produced a large enough sample, which allows us to determine the photometric properties of a representative population of geosats.

The observations presented in this paper were obtained with the U.S. Naval Observatory, Flagstaff Station 40-inch Ritchey telescope on the nights of July 9–11, 2015 UTC. Observations were collected with an LN2-cooled 2048×2048 CCD camera, with pixel size 0.676 *arcsec* (3.28 μ radians), corresponding to a field of view of $\approx 23 \times 23$ *arcmin* (6.7×6.7 m radians). This field of view is large enough to allow us to observe up to 4 geosats at once in fields where multiple targets are present. The availability of multiple targets in the same field allowed us to gauge the intrinsic variability of these targets over a short time scale.

Observations were performed with standard B and V Johnson filters, and R and I Cousins filters (Bessell, 2005). The images were reduced using standard astronomical procedures, including the subtraction of bias frames, the division of the images by a normalized flat-field image observed through the same filter, and the correction of atmospheric extinction using average extinction coefficients measured at USNO. The magnitude calibration was done using observations of a field of standard stars (Landolt, 1992), repeated several times through the night, typically every 30–60 min. Based on these observations we were able to determine that the observing conditions were close to photometric, with little variation between the multiple observations of the calibration field. The average calibration errors were 0.01 mag.

The satellites azimuth and elevation positions, as seen from Flagstaff, were determined based on the most recent Two Line Element sets available at the time of the observations. The satellite observations were performed by pointing the telescope to a target field with the sidereal track disabled. This way the satellites are seen as point sources and the stars appear as streaks in the East-West direction, as shown in Fig. 1. The typical integration times used were 10 s for the I and R bands, 20 s for the V band and 60 s for the B band. These integration times ensured that the shot noise was $< 1\%$, even for the fainter targets. However, integration times as short as half of these values were used for the brightest targets, to avoid saturation. The satellite fields were usually observed using the following sequence of filter: two I band images, followed by two R band, two V band, two B band, with an additional I band image at the end. The bracketing of the observations with an extra I band image was done to gauge the possibility of cloud interference. By comparing satellite fluxes in the first and last I band images, under the assumption that their fluxes don't change by a large amount on a short time scale, we were able to determine that cloud interference was not an issue. Furthermore, by combining this flux information in the case of fields with multiple satellites, we were able to

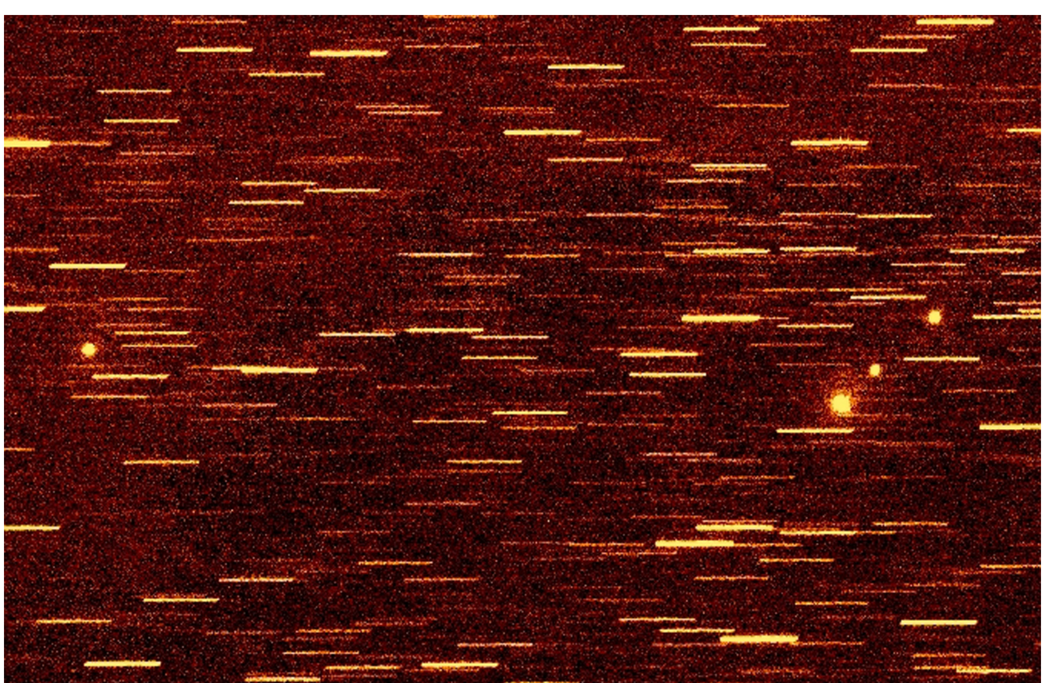

Fig. 1. I band image of a field containing Echostar 17, Anik F1, Anik F1R and Anik G1.

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determine that most sources do not show a large level of variability, usually within the calibration noise, over a ~10 min time scale.

We successfully observed 61 geosats, corresponding to ~66% of our original sample. The remaining 31 satellites were not observed due to the lack of time and instrumental issues. The targets were observed with a solar longitudinal phase angle smaller than ±6°. Throughout our observations the Sun was at a declination of $\sim +22.25°$, resulting in a latitudinal phase angle of $\sim 16.5°$. We use the radiometric phase angle definition, where the phase angle is the measured angular distance between the Sun and the observer, as seen from the target (Africano et al., 2005).

The choice of using a small range of longitudinal phase angles (all observations were done with $|\phi| \leq 6°$, with 70% of the observations centered at $|\phi| \leq 2.5°$, column 8 of Table 1) was an attempt to sample the satellite properties during the period when they are likely to be at their peak brightness. Given that we were only interested in satellites that were active at the time of the observations, this means that their solar panels are oriented in the N-S direction and rotate in order to point directly at the Sun, to maximize power output. Since these observations were done away from the glint season, this approach tends to observe the more diffuse emission from the solar panels. We should also point out that previous studies have shown that one does not expect a large variation in satellite color over the small longitudinal phase angle used for this study, much less over the small range of phase angle over which the observations of a single satellite occured (~12 min, or $\sim 3°$). For example, one can see in Fig. 2 of Payne et al. (2004) that their satellites show a very small color variation (of the order of 0.1 mag or smaller) when observed with a longitudinal phase angle smaller than $\sim 5°$. Similarly, Figs. 5 and 7 of Payne et al. (2009) show multi band photometry of a satellite observed around the winter and summer solstices, similar latitudinal phase angle when compared to our observations. Although the paper does not present colors, one can see in these figures that the different bands trace each other and vary smoothly around $\phi \sim 0°$. Another such example can be seen in Fig. 5 from Sanchez et al. (2000). Even though they only present $J - H$ photometry, their results indicate that in general these colors do not change by more than 0.1 mag around $\phi \sim 0°$.

3. Results

3.1. Sample photometric properties

The results of our observations are shown in Table 1, where we present the names of the satellites, NORAD number, photometry results, number of days since launch at the time of the observations, mean absolute longitudinal phase angle of the observations, manufacturer and bus information.[1] In Fig. 2 we shown the distribution of magnitudes in the different bands for the entire sample. The peak values of the distributions are B = 12.3 mag, V = 11.1 mag, R = 10.5 mag and I = 10 mag. Most of the satellites have magnitudes within ±0.5 mag of the distribution peak, however, we can see a tail of values extending up to ±1.5 mag on either side of the peak, indicating a brightness range of a factor of ~20 (brightest relative to faintest satellite).

The inspection of observed magnitudes presented in Table 1 shows that, even though observations were done over a narrow range of longitudinal phase angles, satellites from the same manufacturer can present a wide range of magnitudes. For example, Loral LS-1300 satellites can have B magnitudes in the range 11.53 mag to 13.37 mag. This range of values can be attributed to differences in the bus and solar panel configurations, a piece of information that is not always available. Furthermore, space weathering can cause the satellite reflectance to change over time (Vrba et al., 2003; Fliegel et al., 2001), reducing the observed flux.

In Fig. 3 we present the distribution of colors (B-V, V-R, R-I and V-I) for our sample. We find that the B-V histogram has a peak around 0.8 mag, with a tail extending towards redder values, B-V = 1.5 mag. The V-R and R-I histograms, show more concentrated distributions, peaked around 0.5 mag, while V-I shows a broader distribution, possibly even double peaked, ranging from 0.5 to 1.5 mag. For comparison, the Sun has B-V = 0.65 mag, V-R = 0.36 mag and V-I = 0.7 mag, which indicates that satellites look redder than the Sun, i.e. significantly brighter at longer than at shorter wavelengths.

3.2. Colors and bus manufacturer

In Figs. 4–7 we present the V-I × B-V, V-R × B-V, V-I × B-R and R-I × V-R color-color diagrams. In these figures we also identify different satellite bus manufacturers with different symbols and colors. These figures confirm that, with a small number of exceptions, satellites are significantly redder than the Sun. We also find that, in general, satellites from the same manufacturer tend to cluster around small regions in these color-color plots, confirming previous results (Payne et al., 2002, 2004). It should be noted that for satellites observed with a different range of phase angles, colors can change significantly, which would change their positions in these color-color plots. Some studies have determined, based on simulations (Payne et al., 2006), laboratory measurement (Cowardin et al., 2010) or observationally for small samples of satellites (Payne et al., 2002, 2004, 2009), how much satellite colors change as a function of phase angle. However, observational results for a sample with a size comparable to the one presented here is not available. Consequently, caution should be exercised when comparing other measurements,

[1] This information was obtained from Gunther's Space Page (https://space.skyrocket.de).

Table 1
Photometry and spacecraft information (Epoch 2015 July 9–11).

Name	Norad	B (mag)	V (mag)	R (mag)	I (mag)	Age[a] (days)	$\|\phi\|$ (deg)	Bus[b]
Star One C2	32768	11.93	10.98	10.24	9.72	2638	3.1	TASB-3000B3
Arsat1	40272	12.31	11.19	10.68	10.29	266	0.8	ARSAT-3K
AMC 6	26580	12.22	11.52	11.10	10.65	5374	0.8	LM A2100AX
Echostar 8	27501	12.60	11.55	11.00	10.50	4704	0.2	LS-1300
Quetzsat 1	37826	12.21	11.15	10.59	9.91	1379	0.2	LS-1300
Echostar 1	23754	11.96	11.26	10.82	10.41	7133	0.2	LM AS-7000
AMC 16	28472	11.59	10.87	10.38	9.99	3856	2.1	LM A2100AXS
XM3	28626	11.82	11.03	10.51	9.93	3782	2.1	BSS-702
Sirius XM 5	37185	12.11	11.01	10.09	9.80	1729	2.1	LS-1300
SES 2	37809	11.14	10.38	9.88	9.54	1387	1.0	Geostar-2.4
Tksat 1	39481	11.88	10.39	9.62	9.16	566	1.0	CGWIC
Spaceway 3	32018	12.09	11.35	10.75	10.12	2886	3.1	BSS-702
Galaxy 3c	27445	12.10	11.23	10.57	9.99	4772	3.1	BSS-702
Intelsat 30	40271	11.86	11.06	10.54	9.98	266	3.1	LS-1300
Echostar 17	38551	11.85	11.00	10.36	9.85	1099	0.1	LS-1300
Anik F1R	28868	13.22	11.92	11.08	10.50	3590	0.1	Eurostar-3000S
Anik F1	26624	10.60	9.68	9.08	8.56	5343	0.1	BSS-702
Anik G1	39127	11.53	10.75	10.22	9.77	814	0.1	LS-1300
XM 4	29520	11.72	10.86	10.30	9.71	3174	1.6	BSS-702
XM 1	26761	10.62	9.82	9.28	8.75	5175	1.6	BSS-702
AMC 7	26495	12.53	11.79	11.28	10.83	5411	1.5	LM A2100A
Goes15	36411	12.05	11.24	10.66	10.12	1953	1.5	BSS-601
AMC 10	28154	12.43	11.78	11.30	10.82	4172	1.5	LM A2100A
AMC 8	26639	12.69	12.07	11.62	11.24	5315	1.8	LM A2100A
Amazonas 3	39078	12.03	10.86	10.11	9.55	882	4.2	LS-1300
Amazonas 2	35942	11.63	10.38	9.64	8.99	2108	4.2	Eurostar-3000
Amazonas 4a	39616	12.28	11.56	11.14	10.74	474	4.2	Geostar-2
Goes 13	29155	11.97	11.22	10.68	10.10	3333	3.6	BSS-601
Star One C3	38991	11.23	10.56	10.17	9.88	1001	0.6	Geostar-2
Nimiq 4	33373	11.64	10.91	10.44	9.97	2485	2.2	Eurostar-3000S
AMC 9	27820	12.07	11.15	10.38	9.74	4415	0.1	ATSB-3000B3
Hispasat 1c	26071	11.81	11.05	10.33	9.73	5635	2.4	ATSB-3000B2
Brasilsat B4	26469	13.77	12.71	12.14	11.77	5439	2.4	HS-376W
Galaxy 17	31307	12.11	11.28	10.57	10.07	3049	0.8	ATSB-3000B3
Nimiq 6	38342	13.37	12.41	11.66	11.07	1148	0.8	LS-1300
Directv 8	28659	12.30	11.51	10.89	10.45	3700	2.2	LS-1300
SES 1	36516	11.40	10.74	10.40	10.09	1902	2.2	Geostar-2
Directv 9S	29494	12.30	11.48	10.87	10.48	3191	2.2	LS-1300
Spaceway 1	28644	12.09	11.33	10.69	10.14	3728	0.1	BSS-702
AMC 1	24315	11.60	10.91	10.46	10.08	6878	0.1	LM A2100A
Directv 12	36131	12.27	11.38	10.76	10.18	2019	4.2	BSS-702
Directv 10	31862	12.35	11.49	10.87	10.26	2925	4.2	BSS-702
SES 3	37748	11.75	10.87	10.41	10.14	1456	4.2	Geostar-2.4
Echostar 11	33207	12.36	11.49	10.90	10.41	2549	2.4	LS-1300
Directv 5	27426	12.56	11.88	11.37	10.91	4811	2.4	LS-1300
Echostar 10	28935	11.13	10.42	9.95	9.53	3431	2.4	LM A2100AXS
Anik F2	28378	12.26	11.37	10.75	10.18	4009	4.7	BSS-702
Wildblue 1	29643	12.44	11.65	11.10	10.68	3135	4.7	LS-1300
Galaxy 23	27854	12.51	11.67	11.13	10.71	4354	0.1	LS-1300
Galaxy 18	32951	12.29	11.54	11.00	10.58	2605	0.6	LS-1300
Galaxy 13	27954	12.43	11.57	10.97	10.46	4298	2.8	BSS-601
Ciel 2	33453	11.72	10.99	10.42	9.94	2402	3.5	TASB-4000C4
Galaxy 12	27715	13.20	12.33	11.87	11.55	4474	3.5	Geostar-2
Echostar 12	27852	10.35	9.65	9.21	8.71	4375	1.0	LM A2100AXS
Echostar 16	39008	12.14	11.25	10.62	10.06	961	1.0	LS-1300
Telstar 14R	37602	12.97	12.11	11.43	10.81	1511	2.5	LS-1300
Nimiq 5	35873	12.74	11.94	11.31	10.78	2121	1.3	LS-1300
Venesat 1	33414	12.53	11.15	10.25	9.46	2444	0.2	CASTC
Intel 16	36397	11.97	11.23	10.87	10.60	1974	1.8	Geostar-2
Galaxy 28	28702	12.12	11.33	10.76	10.24	3668	3.1	LS-1300S
Galaxy 25	24812	12.41	11.71	11.30	10.84	6620	4.2	LS-1300

[a] Number of days since launch.
[b] TASB = Thales Alenia Spacebus; LM = Lockheed Martin; LS = Space System Loral; BSS = Boeing; Eurostar = EADS-Astrium; ATSB = Alcatel Thales Spacebus; Geostar = Orbital ATK; HS = Hughes; CGWIC = China Great Wall Industry Corporation; CASTC = China Aerospace Science Technology Corporation.

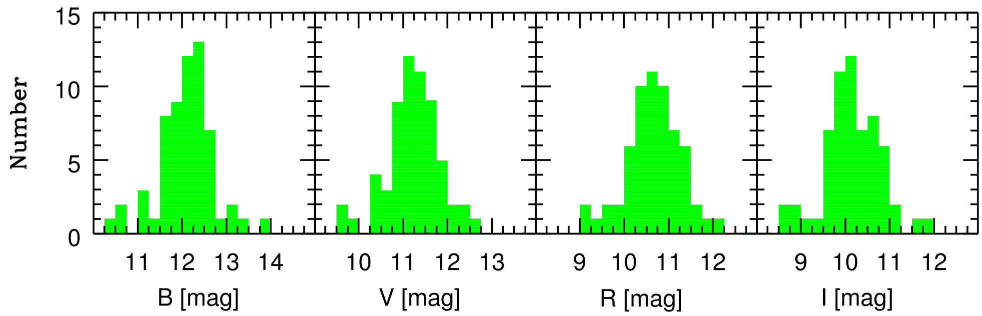

Fig. 2. Distribution of B, V, R and I magnitudes for the sample.

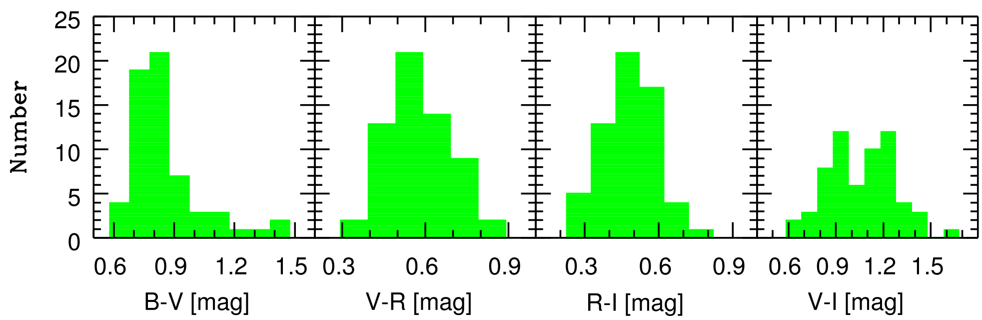

Fig. 3. Distribution of colors for the sample. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

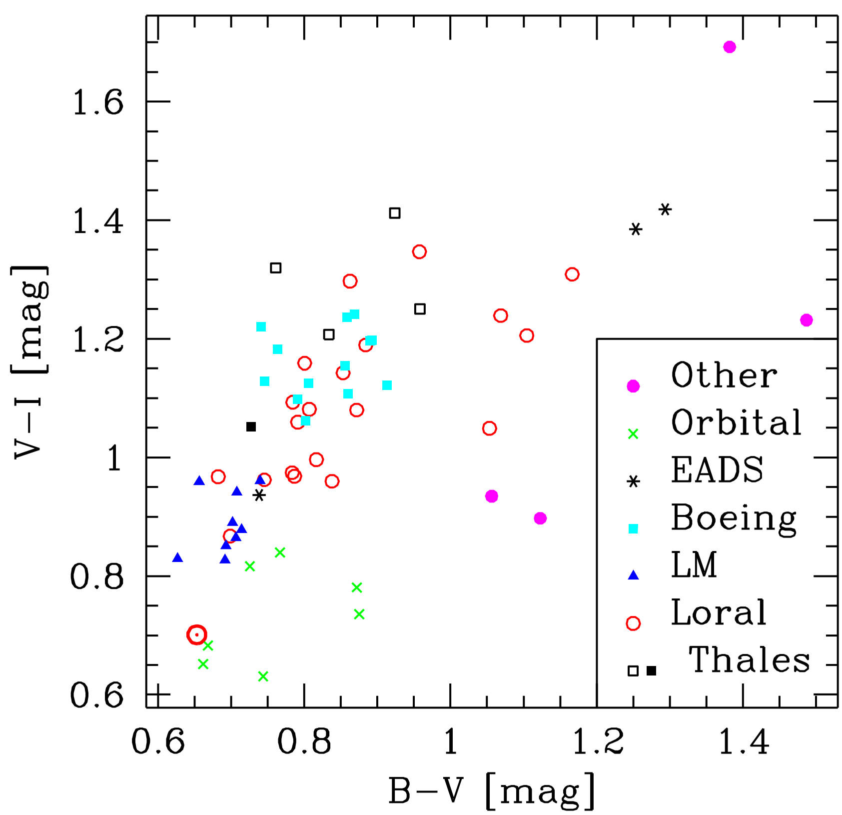

Fig. 4. V-I × B-V plot with the different bus types indicated as different symbols, according to the legend on the right. The symbol ⊙ shows the colors of the Sun. The satellites indicates as Other are Venesat 1, Arsat 1, Tksat 1 and Brasilsat B4. Information about their manufacturers can be obtained in Table 1. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

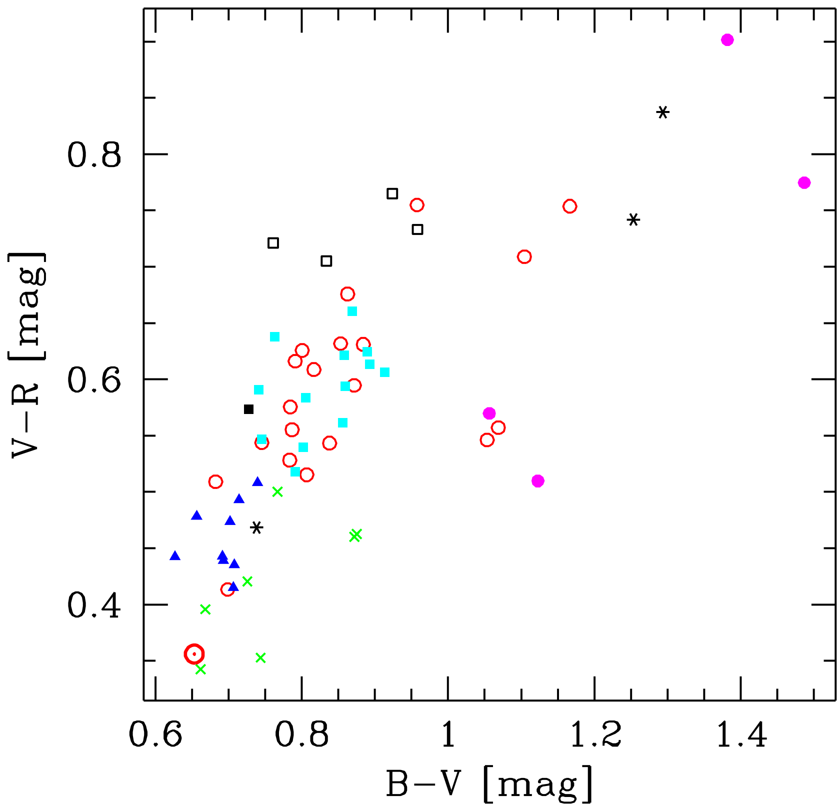

Fig. 5. Distribution of points in the V-R × B-V diagram. Symbols as in Fig. 4.

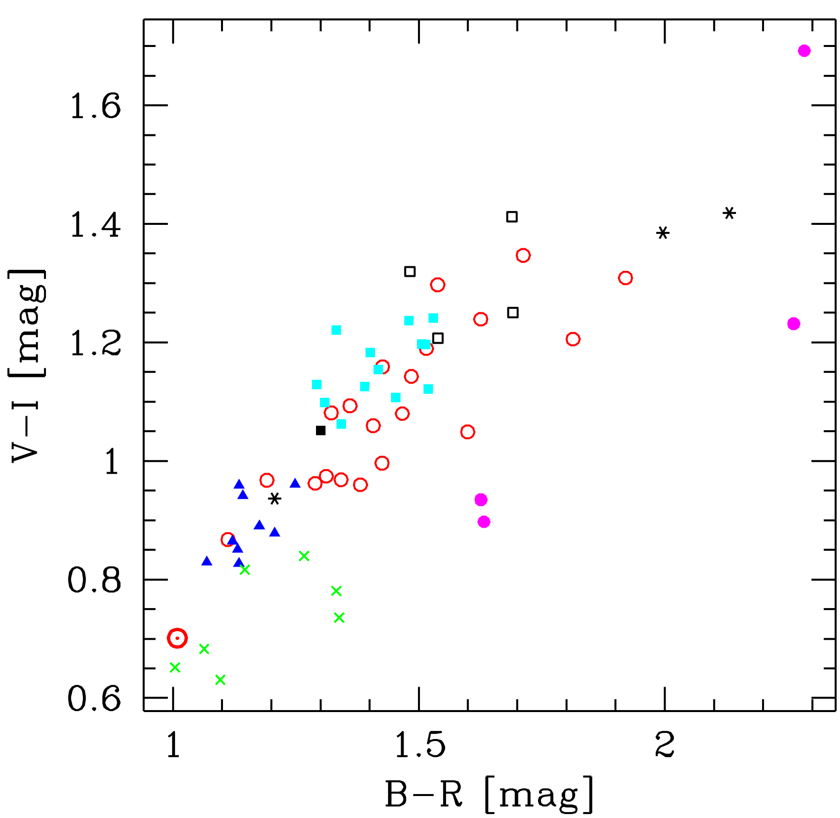

Fig. 6. Distribution of points in the V-I × B-R diagram. Symbols as in Fig. 4.

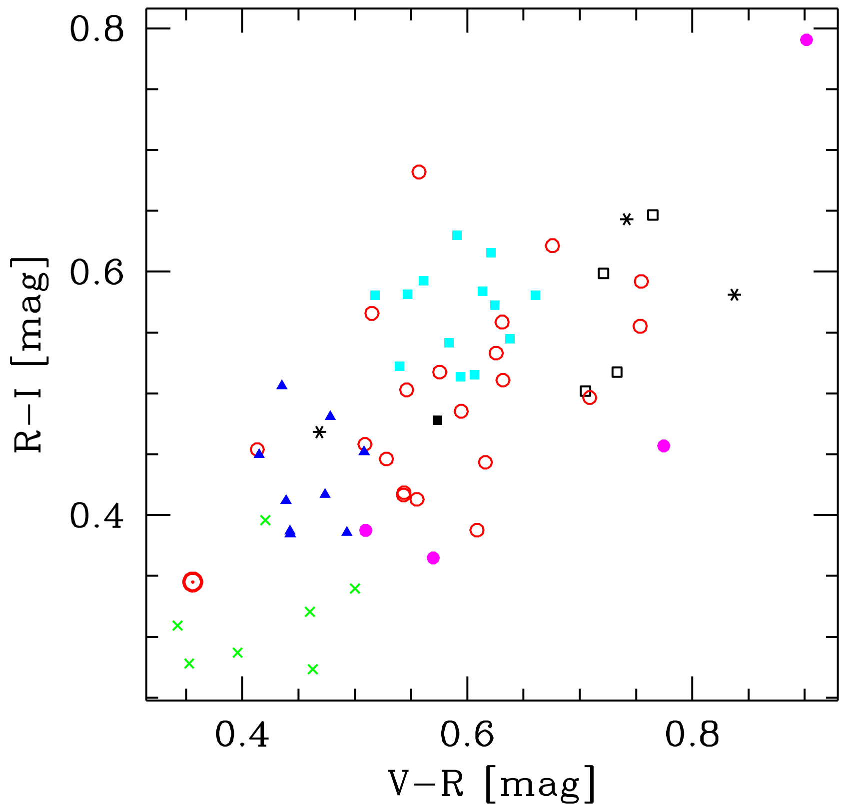

Fig. 7. Distribution of points in the R-I × V-R diagram. Symbols as in Fig. 4.

with a different range of phase angles, with the results presented here.

This clustering of satellites around specific colors is due to the fact that different manufacturers use different materials, as well as different arrangements of the bus components and solar panels, which results in different reflectances as a function of wavelength. A notable exception to this clustering behavior is found for satellites built by Loral, which have colors that spread over a wide range of values, overlapping with clusters from other manufacturers. It is possible that different Loral bus configurations have different colors and cluster along different points in these diagrams, however, this information is not readily available to test this hypothesis. In the case of EADS satellites, for which we have only 3 data points, as well as satellites from 4 different manufacturers for which we have only 1 data point each, we do not have enough information to determine much about their clustering behavior in these color-color diagrams. Nevertheless, 4 of these 7 satellites have the reddest colors in the sample (Tksat 1, Venesat 1, Anik F1R and Amazonas 2), and 2 others (Arsat 1 and Brasilsal B4) have distinctive enough colors, such that they do not tend to overlap with major clusters of other satellites.

A particularly interesting result comes from combining information from Figs. 3 and 4. In Fig. 3 we found that the distribution of B-V values shows a peak around 0.8 and a tail of values extending out to B-V = 1.5 mag. The inspection of the V-I × B-V diagram (Fig. 4) shows that this behavior is due to different manufacturers. The majority of satellites (Boeing, Lockheed Martin, Orbital, Thales-Alenia and most Loral) have B-V < 1 mag, while the ones extending towards larger values are generally related to less common manufacturers, as well as some Loral buses. Similarly, the distribution of V-I values in Fig. 3 shows a double peaked distribution, with peaks centered at 0.9 and 1.2 mag. The inspection of Fig. 4 shows that the first peak (V-I ~ 0.9 mag) is due to Orbital, Lockheed Martin and some Loral satellites, while the second peak (V-I ~ 1.2 mag) corresponds to Boeing, Loral, Thales and other satellite manufacturers.

The results presented in Figs. 4–7 show that multi wavelength photometry is a useful tool in the identification of satellite buses, as well as solving TLE cross-tagging issues, as originally presented by Payne et al. (2002). We should note that Payne et al. (2002) compared results obtained with two sets of filters, a broad band Johnson set similar to the one used here, as well as a SILC set (Space Object Identification In Living Color), designed for satellite classification. They showed that the SILC filters are better at separating between different satellite buses. More recently, results from Xiao-Fen and Yong (2016), who presented a study of 4 satellites using a set of 15 filters, showed the ability to separate between different satellites using photometric results from all filters. They also find that when using a subset of 5 filters they are still able to identify different

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satellites, however, the ability to discriminate between these satellites is diminished. It should be noted that 2 of their satellites had the same bus type and consequently showed very similar colors, consistent with the results presented here. The capability of solving TLE cross-tagging is an important tool in this field. The mean, rms and range of observed colors of satellites from different manufacturers is presented in Table 2.

3.3. Color as a function of age

The database created with the observations presented here can also be used to explore the possibility of determining the age of a spacecraft based on its colors. Observations, as well as Irradiation experiments (Guyote et al., 2006; Abercromby et al., 2007; Engelhart et al., 2017) have found evidence that the interaction of spacecraft materials with high energy particles in space can change the reflectance characteristics of their materials, which can result in a change in their reflectances, and a change in observed colors. The energy, dose and nature of these high energy particles depends on the orbital regime (LEO, MEO or GEO). See Ginet et al. (2013), and references therein, for a detailed study on this subject. Observations of GPS satellites (Fliegel et al., 2001; Vrba et al., 2003) found that these spacecraft become dimmer with time, however, these

Table 2
Mean colors and color ranges for different bus types.

Bus	B-V (mag)	B-R (mag)	B-I (mag)	V-R (mag)	V-I (mag)	R-I (mag)
Thales	0.84 ± 0.100	1.54 ± 0.163	2.09 ± 0.209	0.70 ± 0.074	1.25 ± 0.134	0.55 ± 0.071
	0.72–0.96	1.30–1.69	1.78–2.33	0.57–0.77	1.05–1.42	0.47–0.65
Loral	0.87 ± 0.135	1.46 ± 0.199	1.97 ± 0.248	0.59 ± 0.085	1.10 ± 0.135	0.50 ± 0.077
	0.68–1.17	1.11–1.92	1.56–2.48	0.41–0.76	0.86–1.35	0.38–0.68
Lockheed Martin	0.69 ± 0.033	1.15 ± 0.052	1.58 ± 0.072	0.46 ± 0.031	0.89 ± 0.053	0.43 ± 0.044
	0.62–0.74	1.07–1.25	1.46–1.70	0.42–0.51	0.82–0.96	0.38–0.51
Boeing	0.83 ± 0.058	1.42 ± 0.085	1.99 ± 0.088	0.59 ± 0.041	1.16 ± 0.057	0.57 ± 0.037
	0.74–0.92	1.29–1.53	1.86–2.11	0.51–0.66	1.06–1.24	0.51–0.63
Orbital	0.76 ± 0.087	1.18 ± 0.134	1.49 ± 0.142	0.42 ± 0.059	0.73 ± 0.082	0.31 ± 0.043
	0.66–0.88	1.00–1.34	1.31–1.66	0.34–0.50	0.63–0.84	0.27–0.40

Each entry contains 2 rows. The first row shows the mean and standard deviation of each color for a given satellite manufacturer, while the second row shows the range of values measured for each color.

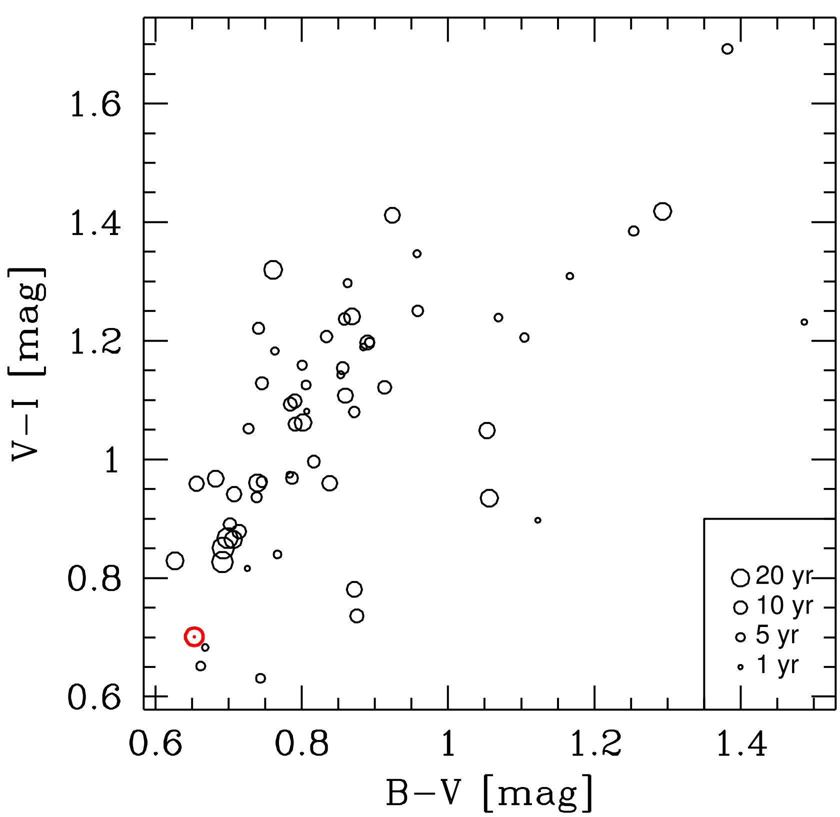

Fig. 8. Distribution of points in the V-I × B-V diagram. The size of the dots shows the bus age, as indicated in the legend.

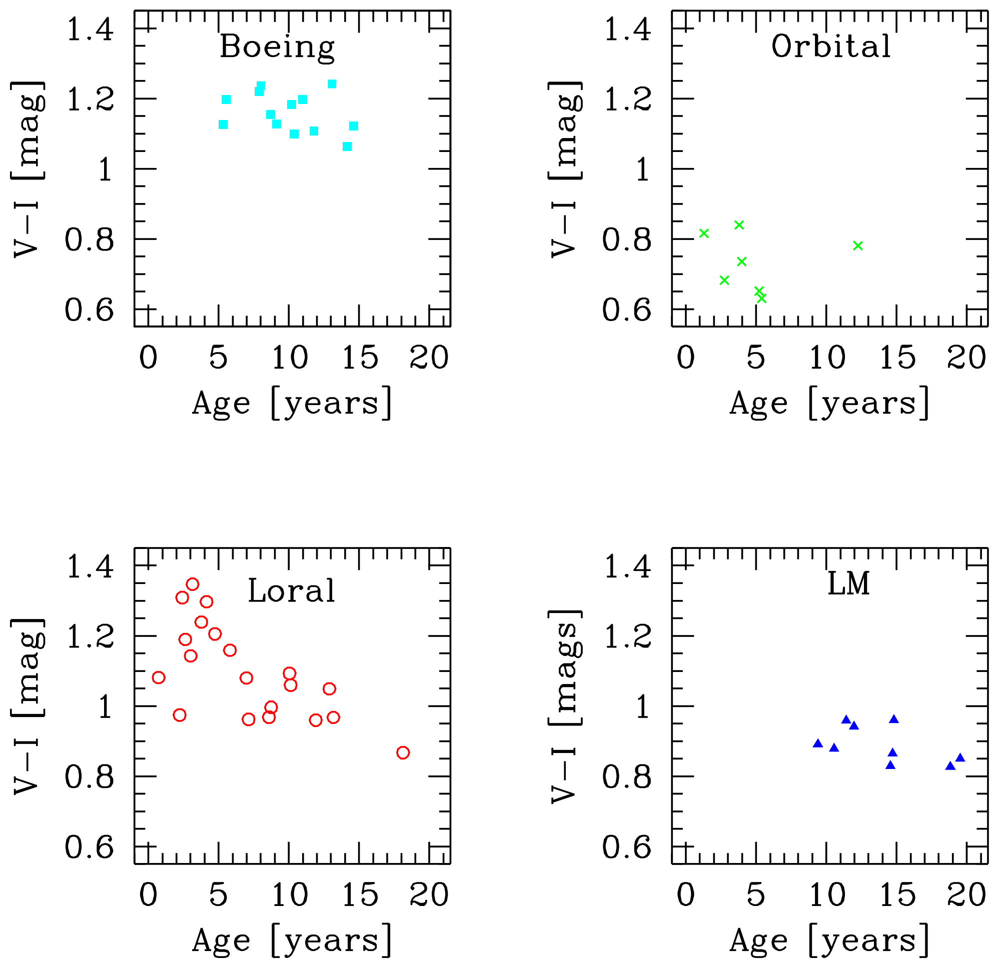

Fig. 9. Distribution of V-I as a function of satellite age, with different manufacturer type shown in a different panel.

authors do not find evidence of an evolution of color (B-V or V-I) with age. In order to explore the possibility of a correlation between satellite age and color, we present in Fig. 8 the V-I $\times$ B-V diagram where the satellite age is encoded in the size of the symbols. This figure shows that in general we do not find a correlation between satellite age and color.

Given that different satellites tend to cluster along different regions in the color-color plots, we investigate whether the lack of correlation between age and color applies to different bus types. In Figs. 9 and 10 we show the distribution of V-I and R-I, respectively, as a function of age, for different satellite manufacturers. We find that in most cases there is no correlation between color and age, with the exception being the Loral satellites, which tend to have smaller (bluer) B-I, V-I and R-I values as a function of age. We find that this negative correlation is statistically significant, with Spearman rank $\rho = -0.54, -0.64$ and -0.56, and probabilities that a correlation is not present $P = 1.9\%$, 0.5% and 1.4% for B-I, V-I and R-I, respectively. For other colors, such as B-V, B-R and V-R, the Loral satellites also show a tendency for smaller/bluer values as a function of age, however, we do not find as strong a statistical significance in these cases ($P > 5\%$). We also find that in the case of the Lockheed-Martin satellites there seems to be a tendency for smaller V-I values as a function of age, however, the range of values is very small and we do not find a high statistically significant correlation. In the case of Boeing and Orbital satellites we do not see any clear correlation between age and color.

The results for most of our satellites agree with those from Fliegel et al. (2001), Vrba et al. (2003)), who did not find an evolution of GPS satellite colors with age, but contradict those from Guyote et al. (2006), Abercromby et al. (2007), Engelhart et al. (2017), who found a general reddening of space objects with age. At this point we can only speculate that the difference between our results, as well as those from Fliegel et al. (2001), Vrba et al. (2003), relative to (Guyote et al., 2006; Abercromby et al., 2007; Engelhart et al., 2017) is due to the orbital environment, since their targets were at LEO.

Although the relation between color and age found for the Loral satellites is interesting, we cannot determine if it is due to weathering of the materials in space, or just a change in the type of materials, or manufacturing procedures over time. Unfortunately the information about manufacturing procedures and materials is proprietary,

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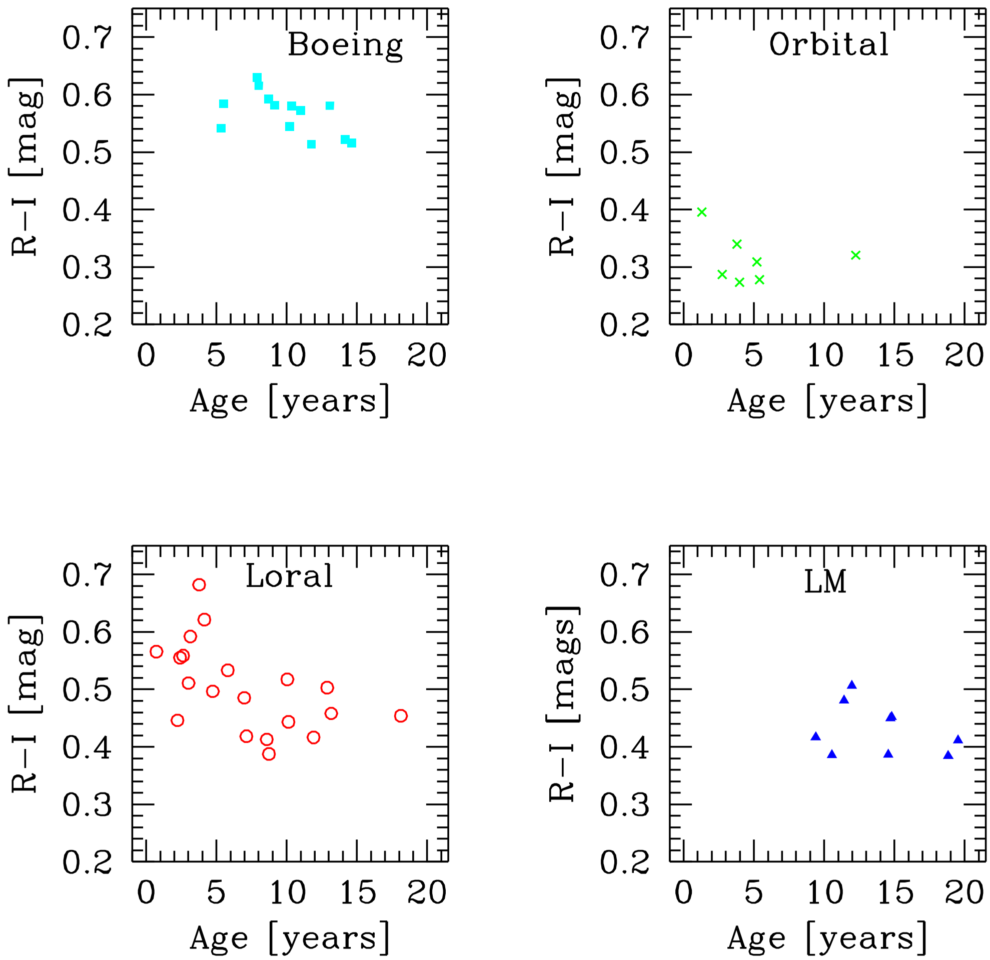

Fig. 10. Distribution of R-I as a function of satellite age, with different manufacturer type shown in a different panel.

not readily available, making it hard to distinguish between these two possibilities. Furthermore, we cannot rule out the possibility that the weathering of the materials happens very fast, over the course of a few weeks or months, since most of the satellites observed in this study have been in orbit for longer than 2 years.

4. Summary

In this paper we presented observations of B, V, R and I photometry of a sample of 61 geosats. All satellites were observed with similar solar phase angles and similar weather conditions. These observations represent an important database for the study of the properties of these targets. This database allowed us to determine the distribution of magnitudes on different bands, as well as the typical colors of these targets. Based on this information we showed that color information is an important tool in the identification of satellites manufacturers, and can play an important role in resolving cross tagged TLEs. We have also used this database to investigate whether a satellite's age can be determined based on broad band colors. We find that in most cases, with the exception of Loral satellites, there is no correlation between color and age. Based on this information alone we are not able to determine if the correlation found for Loral satellites is due to space weathering, or due to a change in manufacturing procedures. In future studies we plan to expand the wavelength coverage and use this database to further study the correlation between the photometric properties and other physical properties of these satellites, such as area.

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